

Section 5.5

I. Differentiation: Chain Rule

A. Examples

II. Integration by Substitution: Reverse the Effects of the Chain Rule

A. Theorem Let $f(x) = g(u(x))u'(x)$ and let G be an anti-derivative of g . Then,

$$\int f(x) dx = \int g(u(x))u'(x) dx = \int G'(u(x))u'(x) dx = \int (G(u(x)))' dx = G(u(x)) + C$$

$$\text{Note: } \int g(u(x))u'(x) dx = \int g(u) du = \int G'(u) du = G(u) + C = G(u(x)) + C$$

III. Examples

A. Indefinite Integrals

B. Definite Integrals

1. Change of Limits